Software and Hardware Tools

1. Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.

* Wireshark Network Traffic Capture

1. Download and install Wireshark on the laptop from its official website.
2. Open Wireshark and select the active network interface, such as Wi-Fi or Ethernet.
3. Click Start Capturing Packets.
4. Open Instagram or YouTube in a browser and browse normally for about 2 minutes.
5. After 2 minutes, click the Stop button in Wireshark.
6. Check the Protocol column to identify the protocols captured.
7. Save the capture using File → Save As and save it as a .pcapng or .pcap file.
8. Note the three protocols that appear most frequently.

2. Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.

```
C:\Users\Prachi Patel>ping www.spotify.com

Pinging atc.spotify.map.fastly.net [2a04:4e42:31::810] with 32 bytes of
data:
Reply from 2a04:4e42:31::810: time=48ms
Reply from 2a04:4e42:31::810: time=40ms
Reply from 2a04:4e42:31::810: time=38ms
Reply from 2a04:4e42:31::810: time=46ms

Ping statistics for 2a04:4e42:31::810:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 38ms, Maximum = 48ms, Average = 43ms

C:\Users\Prachi Patel>ping www.zomato.com

Pinging e70929.dsca.akamaiedge.net [2600:140f:1e00::1737:f4d0] with 32 b
ytes of data:
Reply from 2600:140f:1e00::1737:f4d0: time=38ms
Reply from 2600:140f:1e00::1737:f4d0: time=46ms
Reply from 2600:140f:1e00::1737:f4d0: time=41ms
Reply from 2600:140f:1e00::1737:f4d0: time=50ms

Ping statistics for 2600:140f:1e00::1737:f4d0:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 38ms, Maximum = 50ms, Average = 43ms

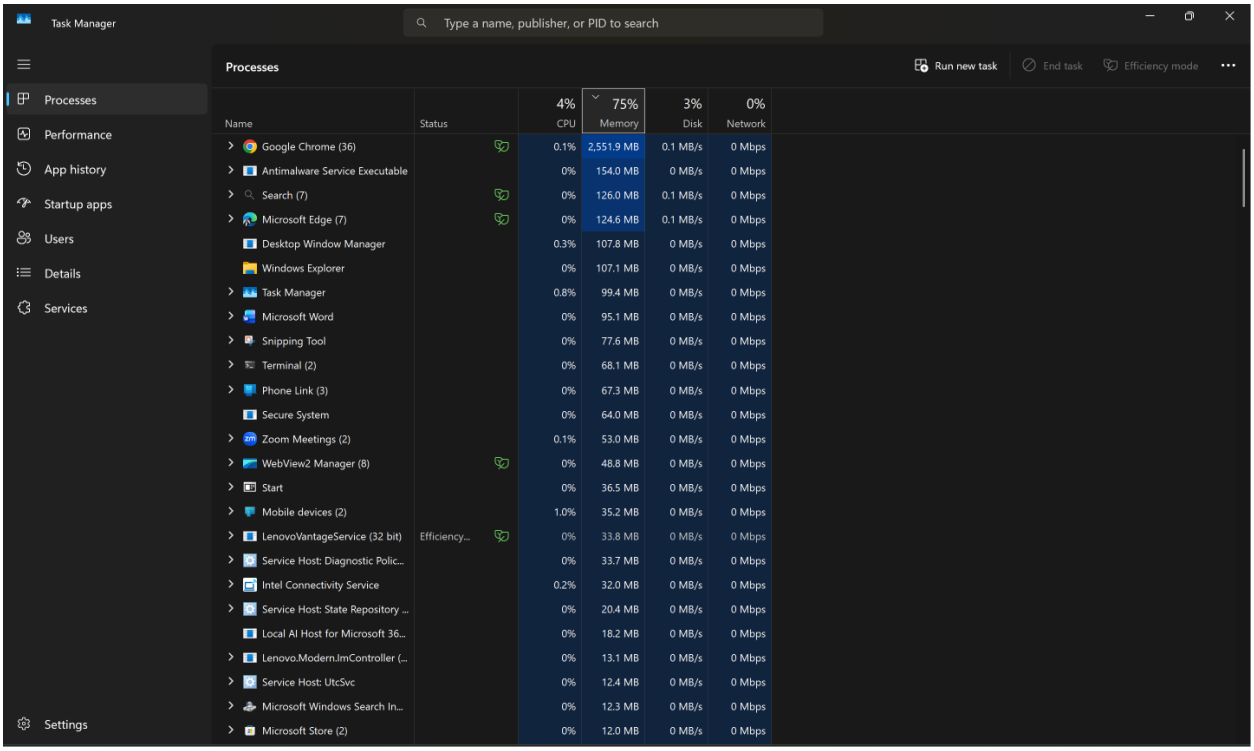
C:\Users\Prachi Patel>
```

I used the ping command to test connectivity to Spotify and Zomato. The response time was measured in milliseconds (ms). The website with the lower average response time responded faster.

Spotify: 43 ms average

Zomato: 43 ms average

3. Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.



**4. Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network.

Hint: Make sure your firewall allows scanning, and only scan your own home/college network.**

A network scanner is used to discover devices connected to a local network. Tools such as Advanced IP Scanner or Angry IP Scanner scan a selected IP address range and display information about the devices that respond.

Steps to perform:

1. Download and install Advanced IP Scanner or Angry IP Scanner.
2. Connect the laptop to your Wi-Fi network.
3. Open the scanner and select the local IP range.
4. Start the scan and wait for it to complete.
5. The scanner will display devices found on the network.
6. Check the IP Address and Device Name/Hostname columns.
7. Record the details of at least two devices.
8. Stop the scan after collecting the required information.

Example:

- Device 1: 192.168.1.5 – Laptop
- Device 2: 192.168.1.8 – Mobile Phone

5. Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.

Hardware Tool	When I would use it
Cable Tester	I would use it to check whether an Ethernet cable is properly connected and identify faults such as broken or incorrectly wired pairs.
Crimping Tool	I would use it to attach an RJ45 connector to the end of an Ethernet cable when making or repairing a network cable.
Multimeter	I would use it to check voltage, continuity, and electrical connections when troubleshooting power-related problems in networking equipment.